B6 Skill Tree Analyses

Modifier Impact Post-mortem + Tier Fitness Assessment

Reincarnated engine — 2026-05-12 — Corpus: 12 B6-populated classes from 2 season(s); 217 classes excluded (B6 fields not populated)

Executive Summary

Corpus statistics: **12 classes** across **9 archetypes**. Mean |modifier – 1.0|: **0.822**. All classes generated under B14.5 V1 primary loop with hybrid rejection gate active.

Headline findings

Analysis 1 — Strongest correlations of tree metrics with |mod – 1.0|:

- **tier_T2**: r = 0.44 (Moderate positive)
- **chain_count**: r = 0.402 (Moderate positive)
- **cross_chain_edges**: r = 0.392 (Weak positive)

Analysis 2 — Tier fitness: Mean role cohesion within tier × archetype: **42.7%.** Mean element cohesion: **82.3%**.

Stratification: 35/35 archetype-tier groups have mean scaling_coefficient inside the design band (T1: 1.05–1.08, T2: 1.08–1.12, T3: 1.12–1.18, T4: 1.18–1.25).

1. Skill Tree Impact on Mean |mod – 1.0|

Analysis 1 correlates per-class skill tree metrics against the class's final balance_modifier distance from 1.0. The hypothesis: certain tree shapes consistently correlate with extreme modifiers, suggesting structural patterns that B14.5 V2 (evaluate-at-converged) or B6 template revisions could target.

Correlations table

Tree Metric	N	Pearson r	Strength	Interpretation
skill_count	12	0.219	Weak	Weak positive
tier_T1	12	0.31	Weak	Weak positive
tier_T2	12	0.44	Moderate	Moderate positive
tier_T3	12	-0.217	Weak	Weak negative
tier_T4	12	-0.279	Weak	Weak negative
chain_count	12	0.402	Moderate	Moderate positive
cross_chain_edges	12	0.392	Weak	Weak positive
total_parent_edges	12	-0.039	Negligible	Negligible negative
mean_energy_cost	12	-0.022	Negligible	Negligible negative
mean_cooldown	12	0.168	Negligible	Negligible positive
mean_damage_mult	12	N/A	N/A	Insufficient data
mean_scaling_coef	12	-0.345	Weak	Weak negative

Key findings

Strongest signal: *tier_T2* shows r = 0.44 (Moderate positive). With current sample size (n = 12), this is suggestive but not conclusive — would benefit from larger B6-populated corpus.

Per-class summary

Class	Archetype	Skills	Chains	|Mod-1.0|	Modifier
class_0005	rogue	13	2	0.5385	0.4615
class_0005	physical_warrior	11	2	0.6828	0.3172
class_0004	wind_controller	13	2	0.757	0.243
class_0007	water_controller	13	2	0.8016	0.1984
class_0003	earth_controller	12	3	0.8387	0.1613
class_0009	wind_controller	13	2	0.8461	0.1539
class_0002	water_controller	13	3	0.8609	0.1391
class_0001	fire_controller	13	2	0.8758	0.1242
class_0004	wind_caster	12	2	0.8758	0.1242
class_0002	hybrid_mage	15	3	0.9203	0.0797
class_0001	hybrid_mage	15	3	0.9203	0.0797
class_0003	earth_caster	11	2	0.9463	0.0537

2. Skill Tree Fitness — the Four Pillars

Analysis 2 examines each archetype's skill tree across the 4 tiers (T1 starters → T4 capstones) for three properties:

Cohesion: within-tier homogeneity. Do skills at the same tier share role/element identity?

Alignment: tier composition matches archetype design intent. Does T1 lean toward starters/primary_attack? Does T4 lean toward capstones/burst_damage?

Stratification: clear progression up the tiers. Mean energy_cost, cooldown_seconds, damage_multiplier should all rise T1 → T4. Mean scaling_coefficient should land in the tier's design band (T1: 1.05–1.08, T2: 1.08–1.12, T3: 1.12–1.18, T4: 1.18–1.25).

Tier × Archetype matrix

Archetype	T	#	Role Dom.	R-Coh%	Elem Dom.	E-Coh%	En	CD	Scale	Band
earth_caster	1	4	utility: 2/4	50	earth: 4/4	100	23	12.5	1.068	Y
earth_caster	2	3	mobility: 1/3	33	earth: 3/3	100	15	6.6	1.101	Y
earth_caster	3	2	mobility: 1/2	50	earth: 2/2	100	21	8.7	1.166	Y
earth_caster	4	2	utility: 1/2	50	water: 1/2	50	16	9.7	1.215	Y
earth_controll	1	3	area_damage: 1/3	33	earth: 3/3	100	21	8.8	1.06	Y
earth_controll	2	3	control: 1/3	33	earth: 3/3	100	16	11.1	1.094	Y
earth_controll	3	3	damage_over_time:	67	earth: 3/3	100	18	4.8	1.14	Y
earth_controll	4	3	utility: 1/3	33	earth: 2/3	67	18	13.2	1.205	Y
fire_controlle	1	5	control: 2/5	40	fire: 5/5	100	21	12.3	1.062	Y
fire_controlle	2	4	damage_over_time:	25	fire: 4/4	100	18	7.8	1.107	Y
fire_controlle	3	2	damage_over_time:	50	earth: 1/2	50	13	7.7	1.142	Y
fire_controlle	4	2	control: 1/2	50	fire: 2/2	100	19	8.6	1.218	Y
hybrid_mage	1	11	utility: 3/11	27	fire: 4/11	36	16	8.8	1.064	Y
hybrid_mage	2	11	burst_damage: 3/11	27	wind: 5/11	45	23	7.4	1.103	Y
hybrid_mage	3	8	area_damage: 3/8	38	earth: 4/8	50	21	6.4	1.139	Y
physical_warri	1	5	area_damage: 3/5	60	physical: 3/5	60	46	5.4	1.067	Y
physical_warri	2	3	burst_damage: 2/3	67	physical: 3/3	100	31	10.5	1.089	Y
physical_warri	3	2	defensive: 1/2	50	physical: 2/2	100	38	9.9	1.149	Y
physical_warri	4	1	primary_attack: 1/	100	physical: 1/1	100	50	0.8	1.211	Y
rogue	1	3	mobility: 1/3	33	physical: 2/3	67	11	9.5	1.069	Y
rogue	2	3	area_damage: 2/3	67	physical: 3/3	100	7	6.1	1.088	Y
rogue	3	4	primary_attack: 1/	25	physical: 3/4	75	11	8.5	1.156	Y
rogue	4	3	defensive: 1/3	33	physical: 3/3	100	12	10.2	1.215	Y
water_controll	1	6	damage_over_time:	33	water: 4/6	67	18	9.6	1.063	Y
water_controll	2	10	area_damage: 2/10	20	water: 9/10	90	19	9.9	1.093	Y
water_controll	3	7	control: 3/7	43	water: 6/7	86	20	13.8	1.158	Y

Archetype	T	#	Role Dom.	R-Coh%	Elem Dom.	E-Coh%	En	CD	Scale	Band
water_controll	4	3	area_damage: 2/3	67	water: 3/3	100	22	9.1	1.225	Y
wind_caster	1	4	burst_damage: 1/4	25	wind: 4/4	100	25	9.7	1.06	Y
wind_caster	2	3	burst_damage: 1/3	33	wind: 3/3	100	23	5.9	1.094	Y
wind_caster	3	2	utility: 1/2	50	fire: 1/2	50	20	9.7	1.169	Y
wind_caster	4	3	mobility: 1/3	33	fire: 2/3	67	19	12.8	1.235	Y
wind_controlle	1	10	mobility: 4/10	40	wind: 7/10	70	18	9.5	1.067	Y
wind_controlle	2	7	control: 3/7	43	wind: 7/7	100	18	10.4	1.1	Y
wind_controlle	3	6	area_damage: 2/6	33	wind: 5/6	83	17	8.9	1.145	Y
wind_controlle	4	3	defensive: 1/3	33	wind: 2/3	67	20	11.9	1.208	Y

Stratification check

All 35 archetype-tier groups land inside design scaling bands. Generator is producing tree shapes that respect the tier-band design intent.

Cohesion observations

- **T1** (n = 9 archetype groups): avg role cohesion 38%, avg element cohesion 78%
- **T2** (n = 9 archetype groups): avg role cohesion 39%, avg element cohesion 93%
- **T3** (n = 9 archetype groups): avg role cohesion 45%, avg element cohesion 77%
- **T4** (n = 8 archetype groups): avg role cohesion 50%, avg element cohesion 81%

3. Gear and Trait Analyses

Gear data is fully populated in the telemetry DB (gear, gear_instances, gear_traits tables) and represents the smart-loot fit-profile architecture in operational form. These analyses validate trait distribution, smart-loot fit semantics, and color cohesion between class identity and gear visual identity.

3.1 Gear distribution by season

Season	Total	Common	Uncommon	Rare	Epic	Legendary	w/ Traits	% w/ Traits	Mean Powe
001005	4378	880	880	880	859	879	1629	37.2%	0.212
001006	400	80	80	80	80	80	158	39.5%	0.223

Across analyzed seasons: **4778 gear instances**, **1787 (37.4%)** carry traits.

3.2 Trait distribution by category × tier

Tier	stat	Total
Common	30	30
Uncommon	20	20
Rare	369	369
Epic	715	715
Legendary	749	749

Category enum maps to trait architecture: *stat* traits = passive stat modifications; *ability* traits = grant new skills/abilities; *granted* traits = automatic effects. The category × tier distribution validates whether higher-rarity gear carries proportionally more valuable trait types — per design intent, legendary gear should skew toward *ability* and *granted* categories.

3.3 Smart-loot fit profile validation

For each B6 class, computed composite fit score across all gear in same season. Composite = mean(energy_match, range_match, role_match) per the smart-loot 70/30 architecture. **Mean fit score across 12 classes: 0.738**. Values close to 1.0 mean gear pool is well-tuned to class loadouts; values near 0.5 indicate room for fit-profile tuning.

Class	Archetype	Energy	Range	Role	Mean Fit	Top Match
class_0001	fire_controlle	mana	medium	control	0.706	off_hand (1.0)
class_0002	hybrid_mage	mana	medium	hybrid	0.749	accessory (1.0)
class_0003	earth_caster	mana	medium	damage	0.78	weapon (1.0)
class_0004	wind_controlle	mana	long	control	0.667	weapon (1.0)
class_0005	rogue	combo	medium	damage	0.744	accessory (0.987)
class_0001	hybrid_mage	mana	long	hybrid	0.737	accessory (0.977)
class_0002	water_controll	mana	medium	control	0.735	off_hand (1.0)
class_0003	earth_controll	mana	medium	control	0.735	off_hand (1.0)
class_0004	wind_caster	mana	medium	damage	0.805	weapon (1.0)
class_0005	physical_warri	rage	close	damage	0.772	weapon (1.0)

Class	Archetype	Energy	Range	Role	Mean Fit	Top Match
class_0007	water_controll	mana	medium	control	0.735	off_hand (1.0)
class_0009	wind_controlle	mana	long	control	0.693	weapon (1.0)

3.4 Color cohesion (class identity ↔ gear visuals)

Color palette overlap (Jaccard-style) between class color_palette and same-season gear color_palettes. **Mean overlap across 12 classes: 0.0%**. Mean of MAX overlaps (best per-class single-gear match): **0.0%**. Low overlap percentages suggest class identity and gear visual identity are loosely coupled — each gets independent color treatment rather than thematic matching. This may be intentional (gear variety) or a coordination opportunity.

4. Findings and Recommendations

Corpus limitations to acknowledge

Current B6-populated corpus is small (**12 classes**) — historical seasons were generated before B6 serialization was wired correctly. 217 classes excluded from analysis. Conclusions are directional rather than statistically rigorous. Larger corpus (20-50+ classes) would strengthen findings substantially. Recommend: regenerate seeds 1005–1007 (and possibly more) once B6 generator path branching is fully understood to bring 30+ classes into analytical scope.

For B6 archetype templates

- The strongest tree-metric/modifier correlation (**tier_T2**: r = 0.44) suggests this dimension of tree shape predicts where modifier converges. Worth investigating whether the B6 archetype template's constraints on this dimension are too loose or too tight.
- Stratification outliers (archetype-tiers with scaling_coefficient outside design bands) are candidate B6 template review items. These represent kits where generator output doesn't match the per-tier power band design intent.

For B14.5 V2 prioritization

- Classes with extreme |mod − 1.0| values are the primary V2 compression targets. Per Analysis 1, tree metrics that correlate with extreme modifiers suggest where lever cycling should focus.
- Low-cohesion tier groups (role or element cohesion below 50%) suggest tree shapes where lever cycling has less reliable signal — composition swaps may produce inconsistent effects across these heterogeneous tiers.

For Unity client / consumer rendering

- All necessary fields for tree rendering are present: *tier* × *chain_id* × *chain_position* for grid layout; *parent_skill_ids* for unlock-line drawing; *color_value* for visual identity; *cross_chain_rule* for connection-eligibility rules.
- Cross-chain edges count is a useful renderer hint — classes with FLEXIBLE rule + many edges have richer tree structure; STRICT classes have cleaner per-chain progressions. Both should render correctly with the existing field set.
- Forward-compat fields (*trait_slots*, *unlock_flavor*, *skill_tree_layout*) are present but unpopulated — Unity work will hydrate these.

Next discrete work items

1. **Diagnose B6 generator path branching** — why earth_caster (and possibly other archetypes) skip B6 field population. Likely a few hours; unblocks larger corpus.
2. **Strengthen schema validation** — assert non-null B6 fields for taxonomy classes (Discipline #8 refinement).
3. **Regenerate seasons 1005–1007** with corrected generator to build proper corpus (~30 min).
4. **Re-run these analyses on larger corpus** — expect substantially stronger statistical signal.

Appendix: Source data

- Seasons analyzed: season_001006, season_001005
- B6-populated classes: 12
- Excluded classes (B6 fields missing): 217

- Excel companion: *b6-skill-tree-analyses.xlsx* (6 tabs with detailed data)